Zilei Shao

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Education

University of California, Los Angeles — Los Angeles, CA
Ph.D. in Computer Science | Advisor: Guy Van den Broeck — *Sep. 2024 - Present*

Harvey Mudd College — Claremont, CA
B.S. in Joint Major of Computer Science and Mathematics | GPA: 3.94/4.0 — *Aug. 2020 - May 2024*

Research Experience

StarAI Lab, Los Angeles, UCLA — May 2024 – Present
Graduate Researcher Advised by Professor Guy Van den Broeck — *Los Angeles, CA*

- Integrating **tractable probabilistic models** into modern neural architectures to improve the training efficiency, inference quality, and controllability of deep generative models.
- Developing **optimization** techniques for probabilistic and generative models, spanning parameter learning, low-variance gradient estimation, and differentiable routing in large-scale architectures.
- Exploring **tokenization** in LLMs as a **neurosymbolic** problem, showing that non-canonical tokenizations of the same input substantially change model behavior.

Uncertainty in Reinforcement Learning, L.A.C.E. Lab, Harvey Mudd College — Jun. 2022 – Jul. 2023
Undergraduate Researcher Advised by Professor Erin Talvitie — *Claremont, CA*

- Developed bounding-box inference to propagate uncertainty in environment, increasing the performance of a model-based reinforcement learning algorithm Model-based Value Expansion (MVE) by over 100%.

Selected Publications

Full publication list

[1] **Zilei Shao***, Ian Li*, Heng Zhao, Kewei Lian, Guy Van den Broeck, Zhe Zeng, Anji Liu. Learning to Couple: Context-Adaptive Joint Denoising for Diffusion Language Models. *Preprint*, 2026. [PDF]

[2] **Zilei Shao**, Anji Liu. Latent State Utilization in Probabilistic Circuits. *Preprint*, 2026. [PDF]

[3] Ian Li*, **Zilei Shao***, Benjie Wang, Rose Yu, Guy Van den Broeck, Anji Liu. Breaking the Factorization Barrier in Diffusion Language Model. *In Proceedings of the 43rd International Conference on Machine Learning, 2026.* (**ICML 2026**) [PDF]

[4] Heng Zhao, **Zilei Shao**, Guy Van den Broeck, Zhe Zeng. ProbMoE: Differentiable Probabilistic Routing for Mixture-of-Experts. *In Proceedings of the 43rd International Conference on Machine Learning, 2026.* (**ICML 2026**) [PDF]

[5] Anji Liu, **Zilei Shao**, Guy Van den Broeck. Rethinking Probabilistic Circuit Parameter Learning. *In Proceedings of the 29th International Conference on Artificial Intelligence and Statistics, 2026.* (**AISTATS 2026 Oral**) [PDF]

[6] **Zilei Shao**, Anji Liu, Guy Van den Broeck. Zero-Variance Gradients for Variational Autoencoders. *In NeurIPS 2025 Workshop on Structured Probabilistic Inference & Generative Modeling*, 2025. (**Best Poster Award**) [PDF]

[7] Renato L. Geh*, **Zilei Shao***, Guy Van den Broeck. Adversarial Tokenization. *In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, 2025. (**ACL 2025 Main Conference**) [Webpage, PDF]

[8] Erin J. Talvitie, **Zilei Shao**, Huiying Li, Jinghan Hu, Jacob Boerma, Rory Zhao, Xintong Wang. Bounding-Box Inference for Error-Aware Model-Based Reinforcement Learning. *Reinforcement Learning Journal, vol. 1, no. 1*, 2024. (**RLC 2024**) [PDF]

Professional Experience

Waymo LLC — Jun. 2025 – Sep. 2025

- Software Engineer Intern in Predictive Planner team.
- Improved motion prediction models by developing dynamic model capacities for more accurate agent forecasting.

Invited Talks

[1] Token Dependencies as Vulnerability and Opportunity in Language Models. *Qualcomm, Sep. 2026; Harvey Mudd College Summer CS Talk, Summer 2026.*

[2] Adversarial Tokenization. *Shang Data Lab, UC San Diego, Apr. 2025.*

[3] Range-to-Range Regression: Propagating Uncertainty in Environment Model. *Harvey Mudd College Summer CS Talk, Summer 2023.*

Teaching Experience

Introduction to Computer Science, UCLA Computer Science Summer Institute Summer 2026

Lead Instructor | Faculty Supervisor: Todd Millstein *Los Angeles, CA*

- Lead instructor for a three-week introductory Python course for high school students with no prior programming background; delivered daily three-hour lectures to a full cohort, designed the entire curriculum, and supervised a team of three teaching assistants.

Teaching Assistant

- CS161 Fundamentals of Artificial Intelligence, UCLA, 2025 Fall, 2026 Fall.
- CS5 Introduction to Computer Science, Harvey Mudd College, 2024 Spring, 2023 Fall.
- MATH179 Mathematics of Big Data, Harvey Mudd College, 2024 Spring.
- CS81 Computability and Logic, Harvey Mudd College, 2023 Spring.
- CS70 Data Structures and Program Development, Harvey Mudd College, 2022 Fall.
- CS181Y CS for Inquiry, Harvey Mudd College, 2022 Spring.
- CS60 Principles of Computer Science, Harvey Mudd College, 2021 Fall.